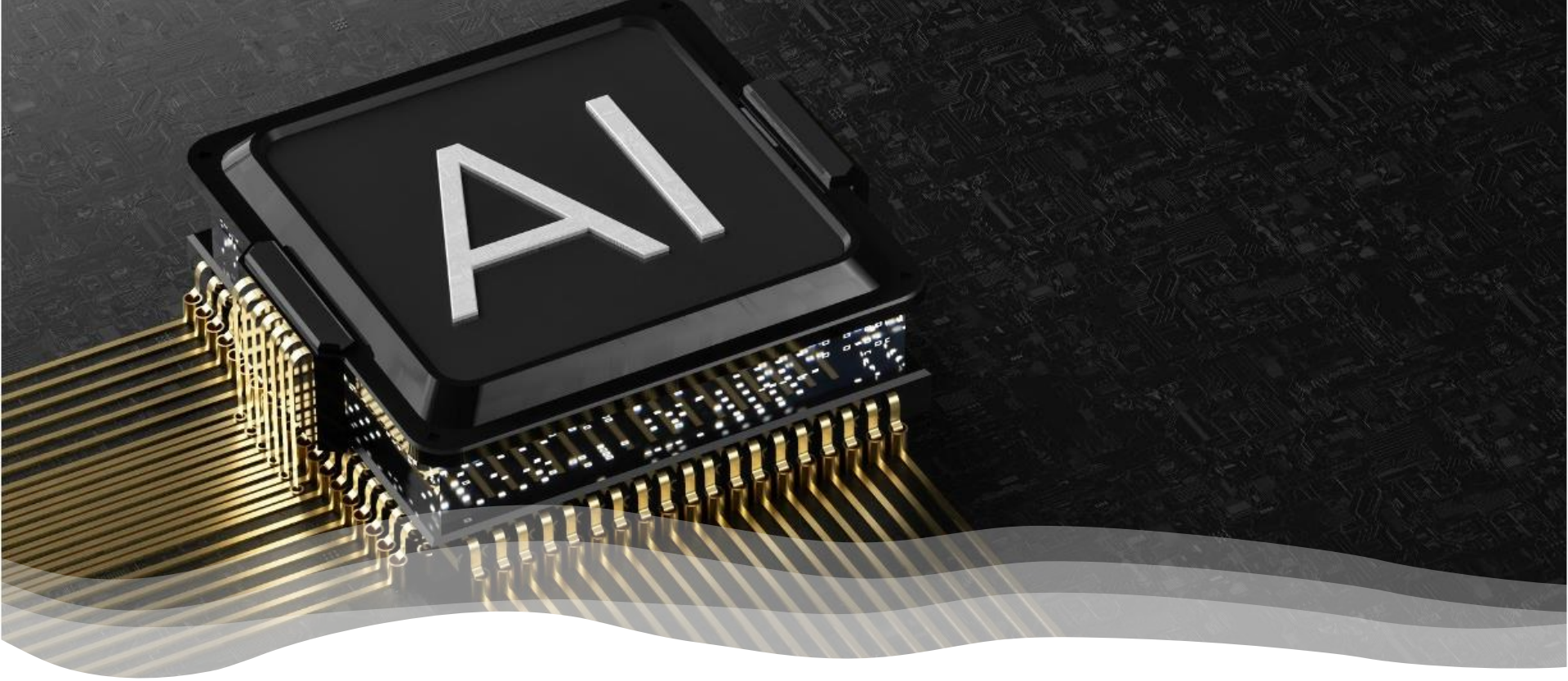
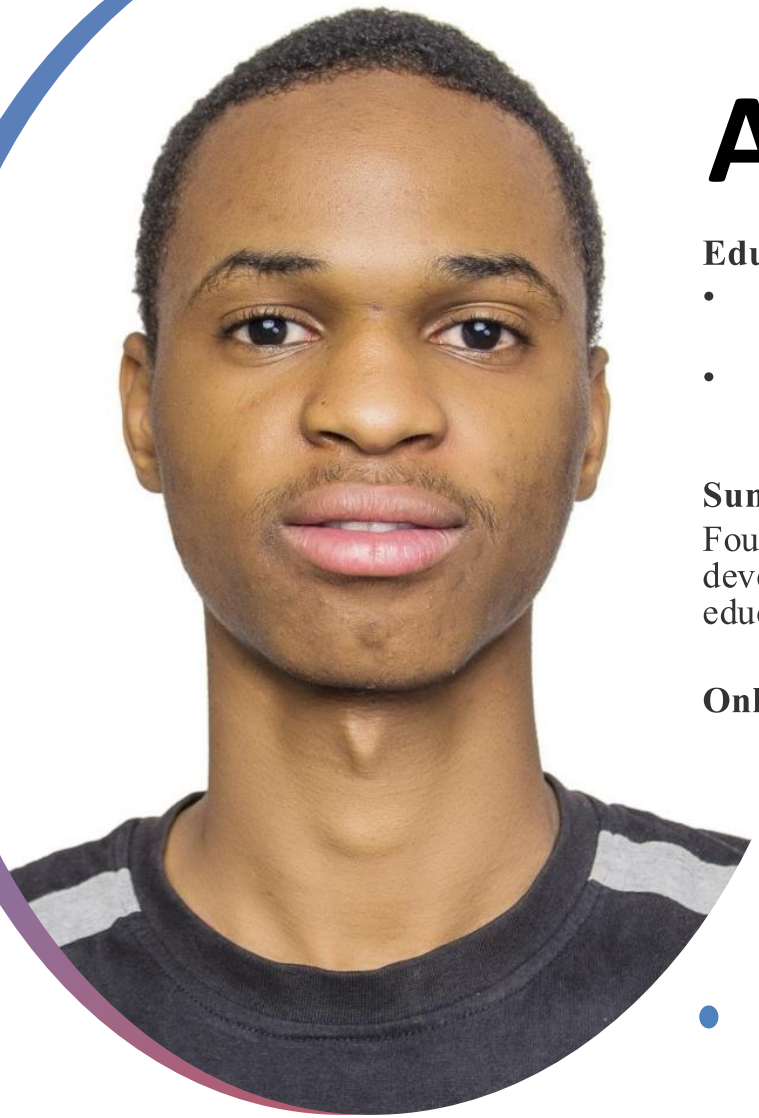


Introduction to AI & Machine Learning
Beginner’s Level



Abubakar Waziri

Education

- MSc in Artificial Intelligence, Georgia Institute of Technology, USA (ongoing)
- BSc in Artificial Intelligence, Prince Mugrin University, Saudi Arabia

Summary

Founder and Chief Executive Officer of AIWaziri, focused on developing AI-driven solutions and digital systems across education, enterprise, and research domains.

Online Presence

GitHub: AiWaziri
LinkedIn: AiWaziri
Website: www.aiwaziri.com
Email:
Phone:

Overview

8-day compressed AI/ML training

Focus:

- Python
- Data
- Machine Learning

Outcome:

- Understand AI/ML basics
- Work with datasets
- Train ML models
- Build a mini AI project (Insha'Allah)

Hardware Requirements

- Laptop or Desktop Computer
- CPU (Intel i3 or higher / AMD equivalent)
- RAM (Minimum 8GB, recommended 16GB)
- Storage (Minimum 20GB free space, SSD recommended)
- Stable Internet Connection
- Optional: GPU (NVIDIA recommended for advanced ML/DL tasks)

Software Requirements

- Operating System: Windows 10/11, macOS, or Linux
- Python (3.8+)
- Jupyter Notebook
- Anaconda / pycharm
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Scikit-learn, tensowrflow etc…
- GitHub (version control)

Daily Structure

- 45 min theory
- 15 min Break
- 1.5 hrs coding
- 30 min review

Day 1: AI Introduction + Python

- AI vs ML vs DL
- Python basics recap
- NumPy fundamentals

Artificial Intelligence (AI)

AI is the broad field of making machines behave like humans:

- thinking
- decision-making
- problem-solving

Example: Chatbots, self-driving cars

Machine Learning (ML)

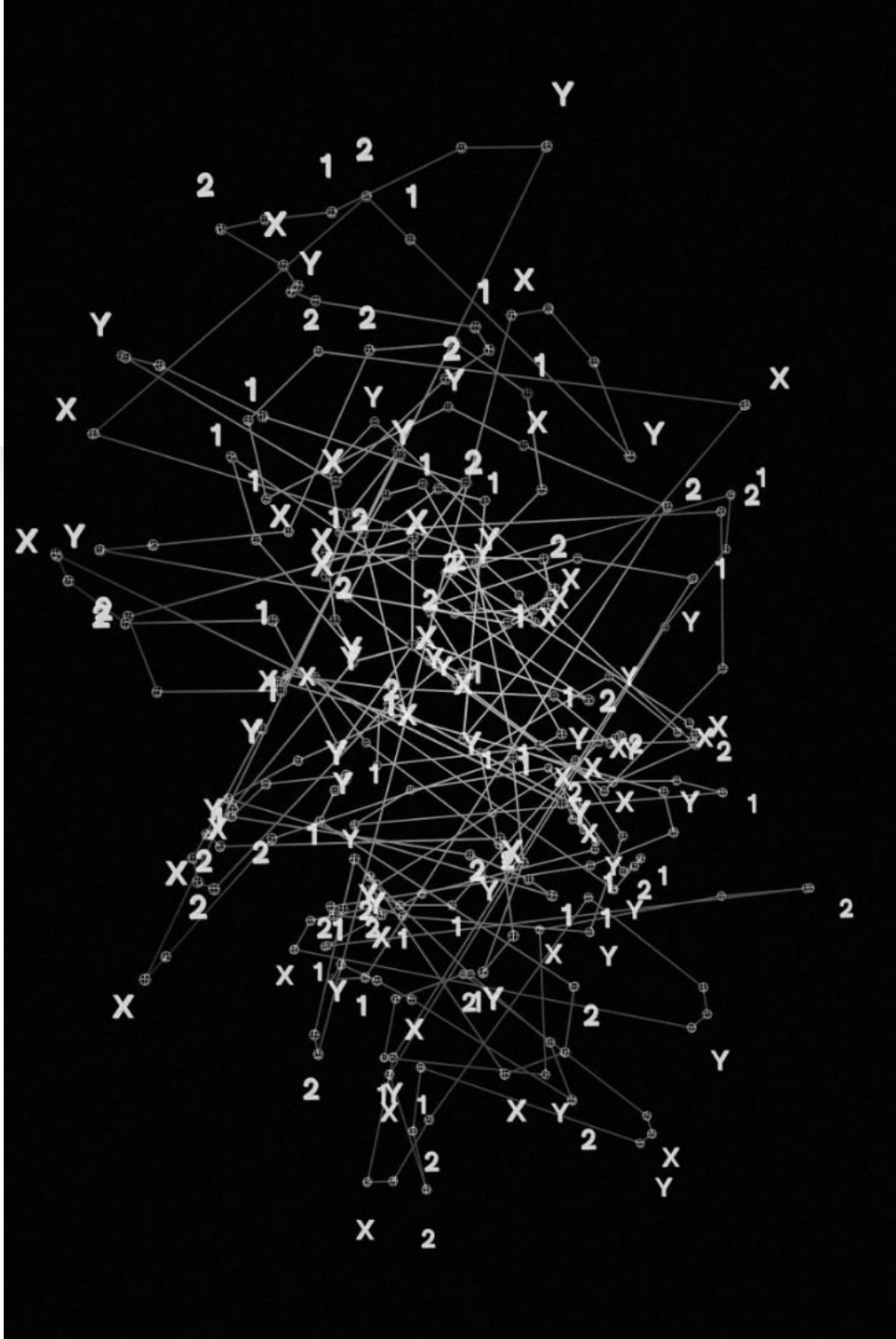
A subset of AI: Machines learn from data without being explicitly programmed

Example: Email spam detection, price prediction

Deep Learning (DL)

A subset of ML: Uses neural networks to learn complex patterns

Example: Face recognition, voice assistants

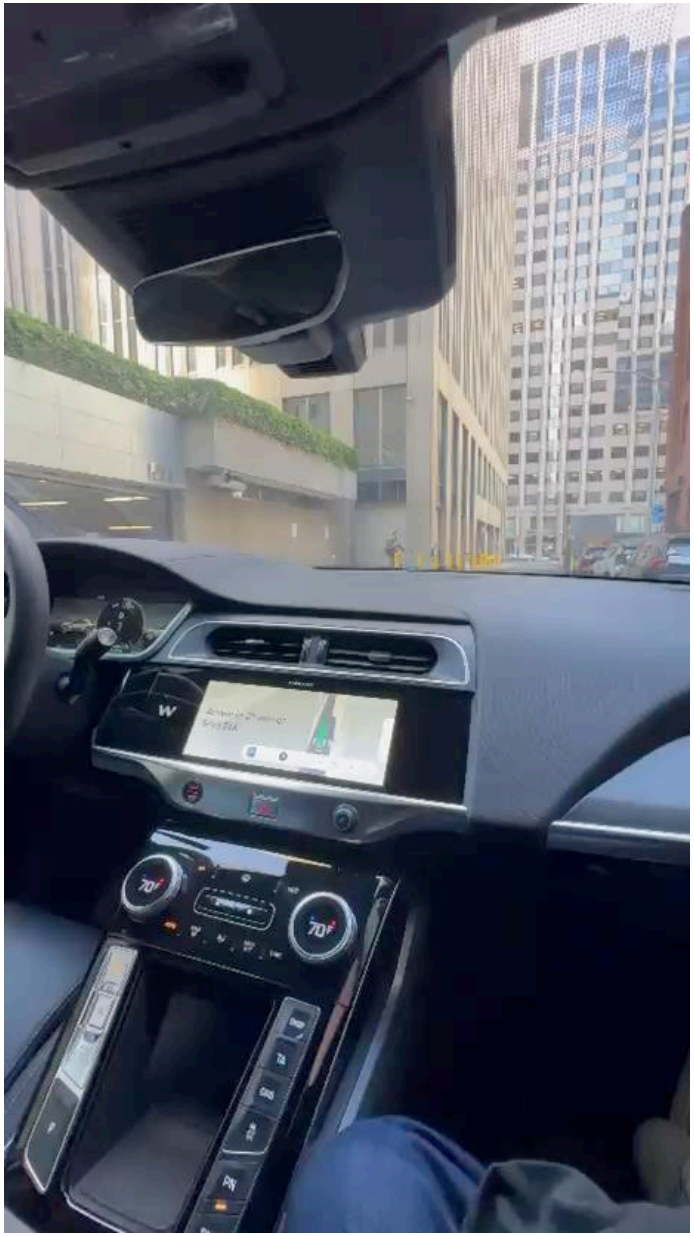


AI vs ML vs DL

Simple Summary

AI ⟶ ML ⟶ DL

Math.math.*



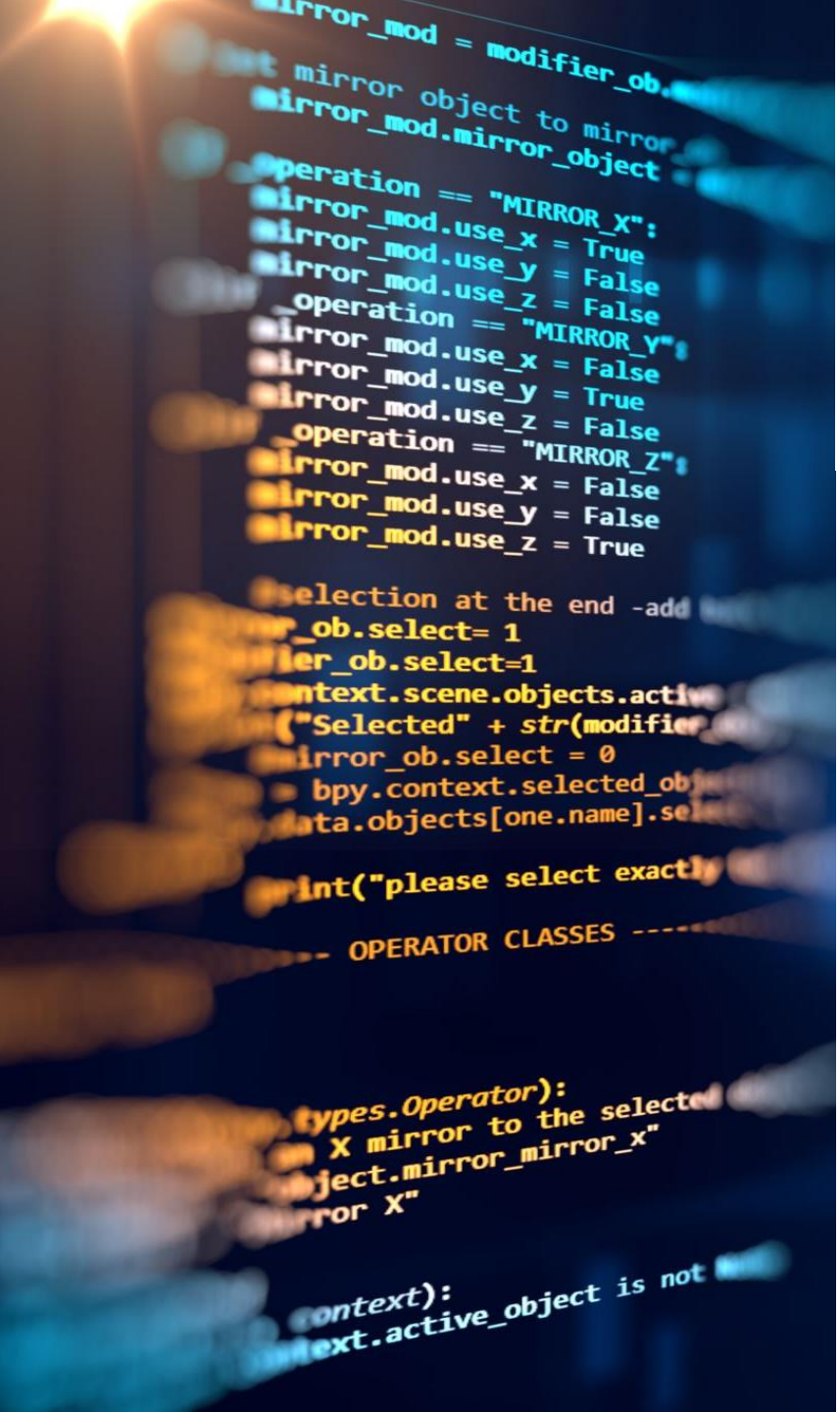
Demo

AI ML & DL:
https://youtu.be/BqtPM4vNzsU?si=NILZcXH9G2JHLRiO

How ai work in daily lives:
https://youtu.be/oJC8VIDSx_Q?si=T6YLnXEX8iUU3HfW

Python basics recap

- **Why Python?**
- Simple and readable language(English)
- Most used language in AI/ML
- OOP



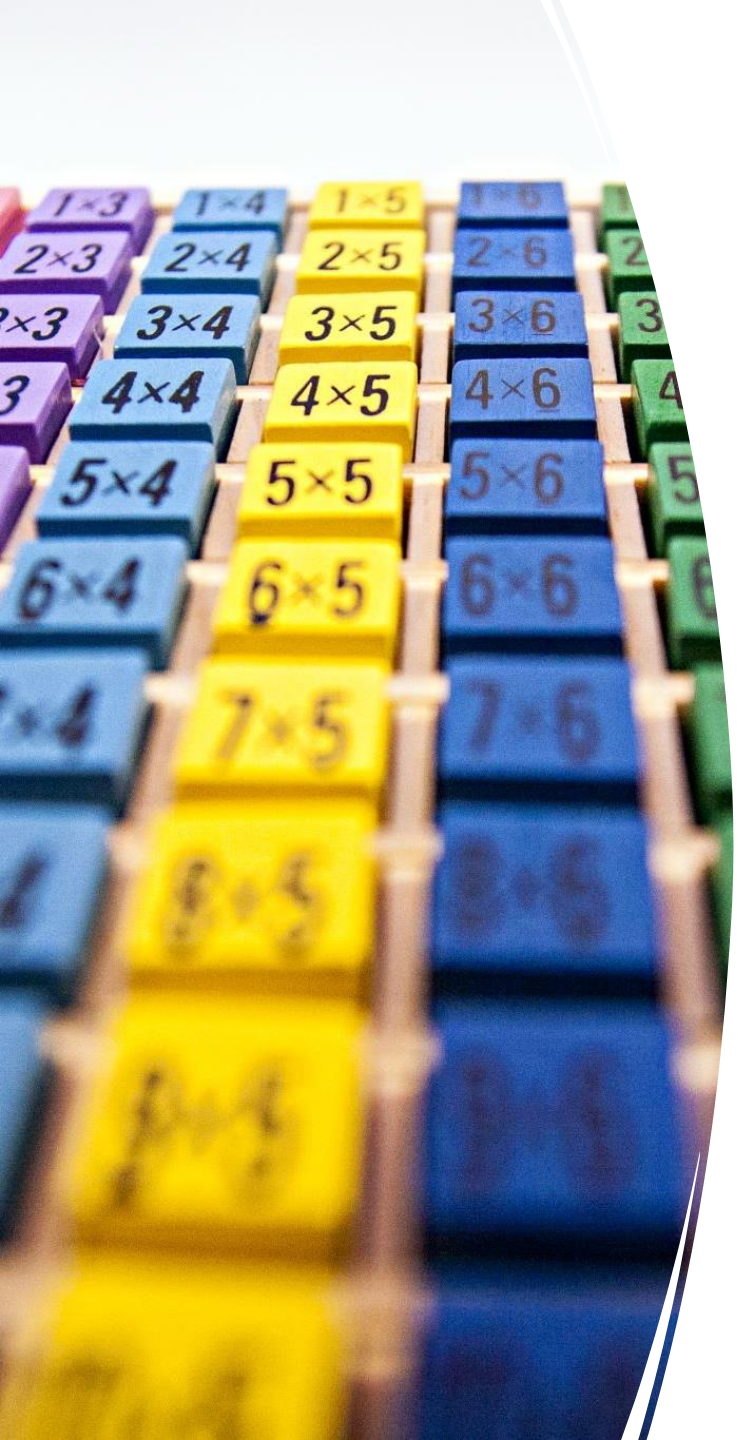
Python basics recap

Key Basics to Remember

- Variables: store data (x = 10)
- Data types: int, float, string, list
- Loops: repeat actions (for, while)
- Functions: reusable blocks of code

Python basics recap

Important Concepts for AI	Lists and dictionaries (for storing data)
	Input/output handling
	Basic logic building (if, while, else, then, until)
Simple Summary	Python is the foundation for writing AI/ML programs.



NumPy fundamentals

- **What is NumPy?**
- A Python library for numerical computing
- Used for handling arrays and mathematical operations

- **Why NumPy is Important**
- Faster than normal Python lists
- Used in almost all AI/ML models

NumPy fundamentals

Key Concepts

- Arrays (1D, 2D, 3D data structures)
- Mathematical operations (sum, mean, multiply)
- Reshaping data

NumPy fundamentals

Basic Example Idea

- Convert list → NumPy array
- Perform fast calculations

Simple Summary

NumPy makes data processing fast and efficient for AI/ML.

Coding Examples

Review + Resources

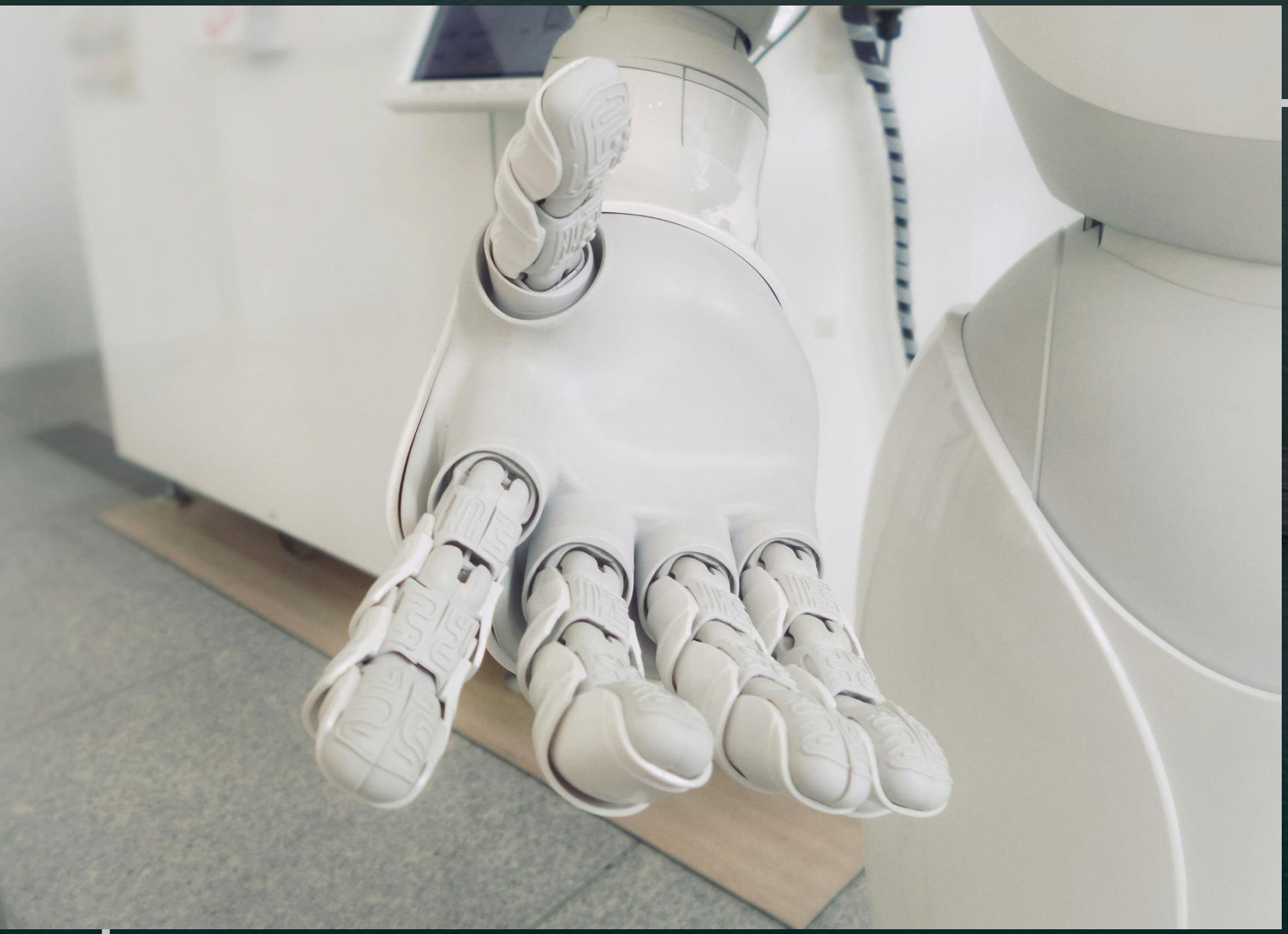
- https://docs.python.org/3/tutorial/
- https://numpy.org/doc/stable/user/quickstart.html
- https://www.w3schools.com/python/numpy/

GitHub materials: https://github.com/AIWAZIRI-LABS/Introduction-to-AI-Machine-Learning-2026-ITng

YouTube:

AI ML & DL: https://youtu.be/BqtPM4vNzsU?si=NILZcXH9G2JHLRiO

How ai work in daily lives: https://youtu.be/oJC8VIDSx_Q?si=T6YLnXEX8iUU3HfW

Neural Network:

- https://youtu.be/rEDzUT3ymw4?si=ai9OLSbErYOFu3W
- https://youtu.be/aircAruvnKk?si=ogsEWZKqg7r-9AAl